(d) What are the requirements of good connector design. Explain the working of expanded beam connector with the help of neat sketch? 7

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4. (a) Define Thermal noise and Shot noise? 2

(b) Describe the working principle of avalanche photo diode. 7

(c) Explain the effect of noise on system performance in photo diode. 7

(d) Derive the expression for responsivity of an intrinsic photo detector in terms of quantum efficiency of the device and wavelength of the incident radiation. Determine the wavelength at which the quantum efficiency and responsively are equal. 7

5. (a) Define attenuation in optical fibres. 2

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(b) Explain the working of optical time domain reflectrometer with help of its block diagram. 7

(c) Compare SONET and SDH optical network. 7

(d) Explain cutback technique for fiber attenuation measurement? 7

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B. E. (Eighth Semester) Examination, Nov.-Dec. 2015

(Old Scheme)

(Et & T Engg. Branch)

OPTICAL COMMUNICATION

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

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Note : *Attempt all questions. Part (a) from each question is compulsory. Attempt any two parts from parts (b), (c) and (d) of each question.*

1. (a) Differentiate between TE and TM mode. 2

(b) When the mean optical power launched into an 8 km length of fiber is 120 μ W, the mean optical power at the fiber output is 3μW. 7

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Determine :

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(i) the overall signal attenuation or loss in dBS through the fiber assuming there are no connectors or splices.

(ii) the signal attenuation per km for the fiber.

(iii) the overall signal attenuation for a 10 km optical link using the same fiber with splices at 1 km intervals each giving an attenuation of 1 dB.

(iv) the numerical input/output power ration in (iii).

(c) What is Soot? Explain the following for optical fiber fabrication : 7

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(i) outside vapour phase oxidation

(ii) inside vapour phase oxidation

(iii) vapour phase axial deposition

(d) Define the following : 7

(i) Snell's law

(ii) Numerical Aperture

(iii) GRIN fiber

(iv) Critical angle

(v) Cladding

(vi) Dispersion

(vii) Scattering

2. (a) Define homojunction and hetrojunction LED. 2

(b) What do you mean by population inversion? Explain the principle of He-Ne gas laser with the help of neat sketch. 7

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(c) Draw the modulation circuits of LED and LASER diode and differentiate them with explanation. 7

(d) A planar LED is fabricated from GaAs which has R.I. equal to 3·6. 7

(i) Calculate the optical power emitted into air as a percentage of the internal optical power for the device when the transmission factor at the crystal air interface is 0·68

(ii) When the optical power generated internally is 50% of the electric power supplied

Determine the external power efficiency.

3. (a) What is fiber pigtail? 2

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(b) Describe the three types of fiber misalignments which may contribute to insertion loss at an optical fiber joint?

(c) What is splicing, explain the following for it : 7

(i) Fusion splicing

(ii) V-grooved splice